

AGML Lab: Week 5

Feature Maps



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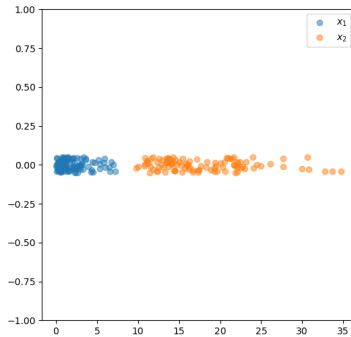
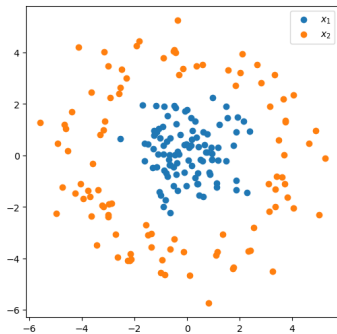
(Jupyter Notebook)

Sometimes, your original feature space is not optimal:

- ▶ not linearly separable,
- ▶ too many features,
- ▶ or even if it is separable, some other separator might still work better



Feature spaces can be used for more than just images:



Einstein summation: a compact representation of tensor operations

Linear Algebra: $u = Av$

unpacked: $\forall i : u_i = \sum_j A_{ij} v_j$

Einsum: $u = \text{einsum}('ij, j \rightarrow i', A, v)$

Imagine we have an image I with RGBA channels and want to halfen the red and blue channel: Let $v = (0.5, 0.5, 1, 1)^\top$, then

Linear Algebra: ???

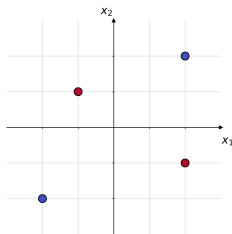
unpacked: $\forall x, y, c : R_{xyc} = I_{xyc} v_c$

Einsum: $R = \text{einsum}('xyc, c \rightarrow xyc', I, v)$

(Jupyter Notebook)

1. Where have we used feature maps before?
2. Find a function that maps

$$X = \begin{pmatrix} 2 & 2 \\ -1 & 1 \\ 2 & -1 \\ -2 & -2 \end{pmatrix}, y = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$



to a space where we can separate the classes with a linear classifier.

3. Create the einsum strings for the following computations:
 - ▶ dot product of two vectors
 - ▶ matrix-matrix multiplication
 - ▶ sum of entries of a vector
 - ▶ trace of a matrix
 - ▶ diagonal matrix